

Memo 10: Concentration-Regime Settling Velocity Formulation for Fine Sediment and MP-Floc Coupling

FVCOM-MP WaterPACT Investigation

May 2026

Purpose

This memo records a replacement direction for the current ORIG_SED plus Stokes-equivalent floc-size workflow. The immediate problem is that a pure Stokes inversion of a capped settling velocity is only a hydraulic diagnostic; it is not a physically consistent floc-size estimate. The alternative reviewed here is the concentration-regime settling velocity formulation used by Shiravani et al. (2023) for microplastic interaction with fine sediment under estuarine conditions. The local reference implementation is `MPTM_FSK_Veloc.m`.

Sources Reviewed

The paper is:

Shiravani, G., Oberrecht, D., Roscher, L., Kernchen, S., Halbach, M., Gerriets, M., Scholz-Bottcher, B.M., Gerdts, G., Badewien, T.H., and Wurpts, A. (2023). Numerical modeling of microplastic interaction with fine sediment under estuarine conditions. *Water Research*, 231, 119564. <https://doi.org/10.1016/j.watres.2022.119564>

ScienceDirect lists the article as open access in *Water Research*, volume 231, article 119564. The paper highlights a new approach for MP settling velocity and describes MP-fine sediment interaction through five settling velocity regions. In the abstract, the key process is described as integration of small MP particles into fine sediment aggregates above a threshold sediment concentration, causing the effective MP settling velocity to approach that of sediment aggregates.

The local MATLAB script `MPTM_FSK_Veloc.m` implements the same concept as a piecewise function of suspended fine sediment concentration. This memo follows the script constants and algebra because it is the version available for direct implementation.

Conceptual Difference from the Present ORIG/Stokes Approach

The current ORIG-based prototype reconstructs a local cohesive settling speed from concentration, salinity, turbulence intensity, and primary sediment size. Then it inverts the settling speed through Stokes law to obtain an “equivalent” floc diameter. This becomes inconsistent when the reconstructed settling speed is capped: many different physical sizes can collapse to the same capped velocity, and Stokes inversion then returns the same small hydraulic diameter.

The concentration-regime formulation avoids that step. It directly prescribes a settling speed as a function of local fine sediment concentration:

- inert region at low sediment concentration;
- transition region where MP/sediment interaction begins;
- flocculation region where settling speed rises with concentration;
- hindered-settling region where settling speed decreases with concentration;
- consolidation region at very high concentration.

This is therefore better suited as a replacement for the effective settling velocity signal. It should not automatically be interpreted as a floc diameter law.

Piecewise Formula in the MATLAB Script

Let C be fine sediment concentration in kg m^{-3} and $w_s(C)$ be the local effective settling speed in m s^{-1} . The script constants are:

Table 1: Constants extracted from `MPTM_FSK_Veloc.m`.

Script name	Meaning in this memo	Value
<code>V0SED</code>	inert MP settling speed	1.67 m d^{-1}
<code>V0SED0</code>	inert MP settling speed	$1.93287 \times 10^{-5} \text{ m s}^{-1}$
<code>Reza</code>	lower interaction threshold C_f	0.1 kg m^{-3}
<code>ch2Sedc</code>	transition/flocculation boundary C_2	1.0 kg m^{-3}
<code>csoil</code>	hindered-settling threshold C_h	8.0 kg m^{-3}
<code>fcgel</code>	gel concentration C_{gel}	50.0 kg m^{-3}
<code>cmax</code>	upper concentration limit C_{max}	250.0 kg m^{-3}
<code>ws0</code>	high-flocculation reference speed	$3.0 \times 10^{-3} \text{ m s}^{-1}$
<code>fgws</code>	gel/consolidation reference speed	$1.0 \times 10^{-5} \text{ m s}^{-1}$
<code>fmfloc</code>	flocculation exponent m	1.1
<code>fbhis</code>	hindered-settling exponent b_h	2.0
<code>cconsol</code>	consolidation exponent β	3.0
<code>wsmin</code>	flocculation-zone lower bound	$1.0 \times 10^{-4} \text{ m s}^{-1}$

The script first computes

$$a_h = \frac{1 - (w_{\text{gel}}/w_{s0})^{1/b_h}}{C_{\text{gel}}}, \quad (1)$$

where $w_{\text{gel}} = \text{fgws}$ and $w_{s0} = \text{ws0}$. It then sets the flocculation coefficient

$$k_1 = \frac{w_{s0} (1 - a_h C_h)^{b_h}}{C_h^m}. \quad (2)$$

For the provided constants, $a_h = 1.88453 \times 10^{-2}$ and $k_1 = 2.19675 \times 10^{-4}$ in SI units.

For the current `V0SED`= 1.67 m d^{-1} case, the script chooses $C_2 = 1.0 \text{ kg m}^{-3}$ and a cubic transition exponent $n = 3$. The transition coefficient is

$$m_R = \frac{k_1 - w_i}{(C_2 - C_f)^n}, \quad (3)$$

where $w_i = \text{VOSED0}$.

The resulting piecewise function is:

$$w_s(C) = \begin{cases} w_i, & 0 < C < C_f, \\ w_i + m_R(C - C_f)^3, & C_f \leq C < C_2, \\ \max(k_1 C^m, w_{\min}), & C_2 \leq C < C_h, \\ w_{s0}(1 - a_h C)^{b_h}, & C_h \leq C < C_{\text{gel}}, \\ w_{\text{gel}} (C/C_{\text{gel}})^{-\beta}, & C_{\text{gel}} \leq C < C_{\text{max}}, \\ 0, & C \leq 0 \text{ or } C \geq C_{\text{max}}. \end{cases} \quad (4)$$

Representative Values

Table 2 shows selected values from the local MATLAB implementation after applying the script constants above.

Table 2: Representative settling speeds from `MPTM_FSK_Veloc.m`.

C (kg m ⁻³)	Regime	w_s (m s ⁻¹)	w_s (m d ⁻¹)
0.01	inert	1.9329×10^{-5}	1.67
0.10	transition start	1.9329×10^{-5}	1.67
0.50	transition	3.6917×10^{-5}	3.19
1.00	flocculation start	2.1968×10^{-4}	18.98
2.00	flocculation	4.7088×10^{-4}	40.68
4.00	flocculation	1.0094×10^{-3}	87.21
8.00	hindered start	2.1636×10^{-3}	186.94
20.0	hindered	1.1647×10^{-3}	100.63
50.0	consolidation start	1.0000×10^{-5}	0.864
100.0	consolidation	1.2500×10^{-6}	0.108

This curve is qualitatively different from the current `ORIG_SED` formula. It does not depend on salinity, local turbulence intensity G , or primary sediment D_{50} . It instead treats concentration regime as the controlling variable.

Updated Modeling Scope

The paper under development focuses on MP-floc interaction, not on resolving floc population dynamics as the main scientific product. Therefore, a full CSTMS/flocmod implementation is not required for the present study if we state the closure carefully. A defensible pathway is to represent the observed estuarine floc field empirically with a small number of cohesive sediment/floc classes and to use the concentration-regime settling formulation as the effective settling-speed closure for MP-floc interaction.

Under this scope:

- the cohesive sediment classes are interpreted as empirical fine sediment/floc classes, not as sand grains;

- class properties and source strengths are tuned so that the simulated turbidity maximum and representative floc-size distribution are reasonable compared with available Delaware estuary observations;
- the model is used to evaluate how plausible MP-floc interaction changes MP transport, water-column partitioning, deposition, and bed storage;
- the work does not claim to mechanistically predict floc generation, aggregation, and fragmentation from first principles.

This keeps the scientific claim centered on MP behavior. Sand-sized median grain classes remain relevant for bed structure, burial, and non-cohesive transport, but they should not be treated as MP-floc aggregate classes.

Implications for FVCOM-MP

The formula directly gives a concentration-dependent settling velocity. That is the quantity we need for:

- differential-settling kernels;
- aggregated MP settling speed;
- output diagnostics such as `settle_vel_floc_mpi`.

However, it does not provide a physical floc diameter. Therefore, using Stokes inversion to recover a diameter would reintroduce the same conceptual problem documented in Memo 09. The next implementation should treat $w_s(C)$ as the primary prognostic/diagnostic coupling signal. If the MP aggregation kernel still requires a size-exclusion diameter, that diameter should be supplied by a separate empirical or parameterized rule, not inferred from pure Stokes inversion of $w_s(C)$.

Implementation Plan

1. Stay within the current ORIG.SED-compatible sediment framework for the next implementation stage. Do not move to CSTMS/flocmod yet. The near-term goal is an empirical multi-class cohesive setup that supports MP-floc sensitivity experiments without claiming full floc-population dynamics.
2. Configure two or three cohesive sediment/floc classes in the sediment input. These classes should represent fine primary material and larger empirical floc/aggregate classes. Their `SED_SD50`, `SED_WSET`, erosion/deposition thresholds, and source strengths should be treated as tunable empirical parameters constrained by Delaware estuary turbidity and floc observations.
3. Add a new optional concentration-regime settling method in `mod_sed.F`, for example `SED_WSET_METHOD = FSK_CONC`. The default must remain the current behavior so existing cases are unchanged. The new method should be available for cohesive classes.
4. Add input parameters corresponding to the MATLAB script constants: `SED_FSK_C_F`, `SED_FSK_C_2`, `SED_FSK_C_H`, `SED_FSK_C_GEL`, `SED_FSK_C_MAX`, `SED_FSK_WS_INERT`, `SED_FSK_WSO`, `SED_FSK_WS_GEL`, `SED_FSK_M_FLOC`, `SED_FSK_B_HIND`, `SED_FSK_BETA_CONSOL`, `SED_FSK_WSMIN`, and `SED_FSK_TRANS_N`. The defaults should reproduce `MPTM_FSK_Veloc.m`.

5. Implement a scalar helper, for example `Calc_Wset_FSK_Cell(C,wset)`, and a column helper, for example `Calc_Wset_FSK(n,c,wset)`. The helper should use only concentration and the configured constants. For multi-class use, the concentration argument should be either total cohesive/floc concentration or class-specific concentration according to a documented switch; the preferred first implementation is total cohesive concentration so the ETM concentration regime controls all floc classes.
6. In the sediment settling routine, branch the cohesive-settling path: retain `Calc_Wset_Floc` for the current ORIG method, and call `Calc_Wset_FSK` when the new method is selected.
7. Expose the same helper to `mod_plast.F` so the MP module can reconstruct the exact same local settling signal from sediment concentration without exchanging `Ws` arrays. This preserves the earlier design principle that MP reads local sediment concentration but does not require a direct settling-speed exchange.
8. Replace the current ORIG/Stokes path in MP with the FSK concentration path for `PLAST_FLOC_EFFECTIVE`. The FSK value should be used directly for differential settling, aggregated MP settling, and `settle_vel_floc_*` output.
9. Use the configured cohesive/floc class diameter directly as the MP aggregation-size variable instead of pure Stokes inversion of `Ws`. This is consistent with the empirical multi-class setup: class diameter represents the modeled floc/aggregate size, while the FSK formula represents the concentration-regime settling speed. A concentration-weighted effective class diameter can be diagnosed for output and plotting.
10. Tune the cohesive/floc source terms and class parameters so that the model produces a plausible ETM and a plausible floc-size enhancement relative to available Delaware estuary measurements. This tuning step should be documented as empirical calibration for MP-floc experiments.
11. Add output diagnostics for the selected concentration regime and the FSK settling speed. Also output the concentration-weighted model-used floc diameter and, where useful, each class contribution to the MP aggregation kernel. Any hydraulic/Stokes-equivalent diameter should be labeled as diagnostic only.
12. Create new paired cases only after the method is implemented: one case with identical multi-class cohesive sediment setup but MP-floc coupling off or using the previous baseline, and one or more cases with FSK MP-floc coupling on for selected MP types. This keeps the sediment environment fixed while evaluating MP response.

Revised Technical Position

The size-exclusion issue should be handled empirically rather than by pure Stokes inversion. The planned multi-class cohesive setup supplies the model-used floc/aggregate diameter from the configured class diameter. The FSK formula supplies concentration-regime settling velocity. These two pieces are therefore separated:

- class diameter controls collision radius, number-density conversion, and the MP size-exclusion rule;
- FSK settling velocity controls differential settling and aggregated MP settling/output;

- source and class-parameter tuning controls whether the ETM produces a realistic empirical floc distribution.

This approach is less mechanistic than CSTMS, but it is appropriate for a paper whose central question is how MP-floc interaction affects selected MP types under plausible Delaware estuary floc conditions.

Implementation Update: Single-Class FSK Case D/E

The immediate implementation will first use a single cohesive sediment class as a formula-isolation test before returning to the broader empirical multi-class setup. Case D and case E will use the same sediment and plastic geometry so that the effect of the MP effective-floc switch can be isolated:

- one cohesive class with `SED_SD50` = 0.010 mm;
- `SED_WSET_METHOD` = `FSK_CONC`, where the inert settling speed is computed internally from `SED_SD50` using the same temperature-compatible Stokes form used by the MP effective-size inversion;
- a microfibre-like MP with `PLAST_AAXI` = 0.010 mm, `PLAST_BAXI` = 0.002 mm, and `PLAST_CAXI` = 0.002 mm;
- case D sets `PLAST_FLOC_EFFECTIVE` = T;
- case E sets `PLAST_FLOC_EFFECTIVE` = F.

This single-class test intentionally keeps the Stokes-equivalent inversion in the MP effective path. Sediment computes the FSK concentration-regime settling velocity, and the MP module reconstructs the same FSK settling velocity from local sediment concentration, sediment class properties, local water density, and temperature. The MP module then applies the existing Stokes inversion with `PLAST_FLOC_RHO_EFF` to obtain the effective floc diameter used by the current MP-floc interaction parameterization. This is a diagnostic implementation test of the FSK settling pathway, not the final claim that FSK settling uniquely determines physical floc size.

Wu et al. Minimum-Dimension Incorporation Criterion

The MP-floc incorporation threshold will be revised to match the size definition used by Wu et al. (2024), *Flocs as vectors for microplastics in the aquatic environment*. In that work, MP size for the boundary curve is the smallest relevant particle dimension: diameter for microspheres, diameter for microfibres, and minimum Feret diameter for irregular fragments. Therefore, the model should not use the volume-equivalent diameter `dequ` when testing whether an MP can be incorporated into a floc.

The code-design split is:

- `dequ` remains the size used for MP settling, collision radius, mass, and volume-related calculations;
- the incorporation threshold uses `min(aaxis, baxis, caxis)`, which is normally `caxis` under the input convention;

- the same minimum-dimension value is used in both aggregation size-exclusion and disaggregation threshold calculations.

For the planned 10 μm floc test, the Wu threshold gives $d_{\text{MP,max}} \approx 3.58 \mu\text{m}$. The selected 10 by 2 by 2 μm microfibre has a minimum dimension of 2 μm and should therefore pass the incorporation gate, while retaining its non-spherical geometry in the rest of the MP physics.